

Bezout's theorem

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In this paper we will define some basic concepts in algebraic geometry such as Affine and Projective spaces as well as mention some pivotal Lemmas that will not be proven culminating in the proof of Bezout's Theorem. The statement of Bezout's Theorem that will be proven in this paper is:

Theorem: Suppose $\mathbb{F}$ is a field and P and Q are polynomials in $\mathbb{F}[x, y]$ with no common factor of degree ≥ 1 . Let $Z(P, Q) := \{(x, y) \in \mathbb{F}^2 \mid P(x, y) = Q(x, y) = 0\}$. Then the number of points in $|Z(P, Q)| \leq \deg(P)\deg(Q)$.

This theorem has a more general form that will be mentioned at the end that is defined over projective space and requires Schemes to prove. In addition, this paper will define Schemes as well as projective space. This is done to provide a better understanding of this simplified version of Bezout's Theorem, but will ultimately be left unproven in this more general form.

To start we will consider the field $\mathbb{F}$ and recall a field is a set $\mathbb{F}$ together with two commutative binary operations $+$ and $\bullet$ on $\mathbb{F}$ such that $(\mathbb{F}, +)$ is an abelian group with identity 0 and $(\mathbb{F} - \{0\}, \bullet)$ is an abelian group, coupled such that distributivity holds. So

$$a \bullet (b + c) = (a \bullet b) + (a \bullet c), \forall a, b, c \in \mathbb{F}$$

Now moving forward recall $\mathbb{F}[x, y]$ is the set of polynomials of two variables. That is if $P \in \mathbb{F}[x, y]$ then

$$P = \sum_{i=0}^n \sum_{j=0}^m a_{ij} x^i y^j$$

with $\deg(P) = n + m$ where $a_{nm} \neq 0$.

The significance of Bezout's Theorem is that in this simplified form we get a bound for the number of intersections of two polynomials on a domain, which is huge as polynomials can be used to represent and agree with a large number of curves on a domain, thus we have a way of bounding the number of intersections of two curves with polynomial representations on a given domain. In the more generalized version of Bezout's Theorem we will find that this bound is actually not a bound but instead a count, in that the bound is the total number of intersections. Thus

the geometric implications of Bezout's Theorem restrict the geometry of two curves by the degree of their polynomial representations.

Lets look at some examples:

First consider polynomials $P : \mathbb{F} \rightarrow \mathbb{C}$ and $Q : \mathbb{F} \rightarrow \mathbb{C}$ given by $P = x - y$ and $Q = x - y^2$ respectively. Then $Z(P, Q) = \{(0, 0), (1, 1)\}$ as $P(x, y) = 0 \iff x = y$ and $Q(x, y) = 0 \iff x = y^2$, so this system of equations has two solutions $(0, 0)$ and $(1, 1)$. Thus the order or $|Z(P, Q)| = 2$, and the $\deg(P) = 1$ and the $\deg(Q) = 2$ so $\deg(P)\deg(Q) = 1 \times 2 = 2$. And as Bezout's Theorem states $|Z(P, Q)| \leq \deg(P)\deg(Q)$ and $2 \leq 2$ so Bezout's holds.

The second example we will consider is $P : \mathbb{F} \rightarrow \mathbb{R}$ given by $P = x^2y^3 - 5$ and $Q : \mathbb{F} \rightarrow \mathbb{R}$ given by $Q = x^2 - y$. Then solving this system we find

$$x^2y^3 = 5$$

$$\implies x^2 = \frac{5}{y^3}$$

and as

$$y = x^2$$

$$\implies y = \frac{5}{y^3}$$

so we have

$$y = \pm \sqrt[4]{5}$$

And as we are only concerned with real solutions, we have

$$x^2 = \frac{5}{y^3}$$

$$\implies x = \pm \sqrt[8]{5}$$

so we have two points in $Z(P, Q) = \{(\frac{1}{\sqrt[4]{5}}, \sqrt[4]{5}), (-\frac{1}{\sqrt[4]{5}}, \sqrt[4]{5})\}$ Thus we have $|Z(P, Q)| = 2$ and $\deg(P) = 5$, $\deg(Q) = 2$ so $\deg(P)\deg(Q) = 5 \times 2 = 10$ and certainly $2 \leq 10$ so Bezout's Theorem holds as $|Z(P, Q)| \leq \deg(P)\deg(Q)$.

Proof:

Suppose $\mathbb{F}$ is a field and P and Q are polynomials in $\mathbb{F}[x, y]$ with no common factor of degree ≥ 1 . Let $\deg(P) = n$ and $\deg(Q) = m$ so we have

$$P = \sum_{i=0}^{l_P} \sum_{j=0}^{k_P} a_{ij} x^i y^j$$

$$Q = \sum_{i=0}^{l_Q} \sum_{j=0}^{k_Q} b_{ij} x^i y^j$$

where $l_P + k_P = n$ and $l_Q + k_Q = m$.

Then look at the form

$$P(x, y) = a_0(x)y^r + \dots + a_r(x)$$

$$Q(x, y) = b_0(x)y^s + \dots + b_s(x)$$

Where $R(x) = \text{Res}_y(P, Q) \in \mathbb{C}[x]$ so we have

$$R(x) = 0 \iff P(x, y), Q(x, y) \in \mathbb{C}[y]$$

and since P and Q share no common roots

$$\deg_x R \leq \deg_x P s + \deg_x Q r$$

so it follows

$$\deg_x R \leq mn$$

and by similar logic we have Where $S(y) = \text{Res}_x(P, Q) \in \mathbb{C}[y]$ so we have

$$S(y) = 0 \iff P(x, y), Q(x, y) \in \mathbb{C}[x]$$

and since P and Q share no common roots

$$\deg_y S \leq \deg_y P + \deg_y Q$$

so it follows

$$\deg_y S \leq mn$$

So it must be true

$$\deg(P \bullet Q) \leq mn$$

Thus,

$$|Z(P, Q)| \leq \deg(P)\deg(Q)$$

Proving Bezout's Theorem.

QED

This concludes the portion of this paper dedicated to the simplified version of Bezout's Theorem. To conclude, the general statement of Bezout's Theorem.

We start by looking at the general projective space we must first define affine space. Suppose $\mathbb{F}$ is a fixed algebraically closed field. Then affine n -space over $\mathbb{F}$ is the set of n -tuples $(x_1, \dots, x_n)$ where $x_i \in \mathbb{F}$, denoted by $A_n(\mathbb{F})$. Note if $\mathbb{F}$ is the field of real numbers $\mathbb{R}$ then $A_n(\mathbb{F}) = A_n(\mathbb{R})$ can be considered euclidean n -space absent of mention of distance. Projective space is then defined as the set of equivalence classes $(n + 1)$ -tuples $(x_1, \dots, x_n)$ with $x_i \in \mathbb{F}$ such that not all x_i are zero under the relation $(x_1, \dots, x_n) \sim (\lambda x_1, \dots, \lambda x_n), \forall \lambda \in \mathbb{F}, \lambda \neq 0$.

So as both affine space and projective space require $\mathbb{F}$ to be a fixed algebraically closed field in the general case it follows in our simplified case we have that $\mathbb{F}$ is a field.

To wrap up definitions we define an affine scheme to be a locally ringed space isomorphic to $\text{Spec}(R)$ where R is a commutative ring and $\text{Spec}(R)$ is the spectrum of R . A scheme is a locally ringed space X with open covering by sets U_i where each U_i is an affine scheme. More X comes with a sheaf $\mathcal{O}_X$ which assigns every open subset U a

commutative ring $\mathcal{O}_X(U)$ called a regular ring of functions on U .

Theorem: Suppose X is a projective subscheme of $\mathbb{P}^n$ of positive dimension. Let $f \in \mathbb{F}[x_1, \dots, x_n]$ be a homogeneous polynomial such that no component of X is in $Z(f)$. Then

$$\deg(X \cap Z(f)) = \deg(X)\deg(f)$$